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324311 (14)

BE (3rd Semester)

Examination, April-May, 2014

Branch : Electrical, EEE

MATHEMATICS - III

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part 'a' is compulsory.

(ii) Attempt any two parts from each unit.

Q. 1. (a) State Dirichlet's condition. 2

(b) Expand $f(x) = x \sin x$, $0 < x < 2\pi$ in a Fourier series. 7

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(c) Obtain a half-range cosine series for f(x)

kx, $0 \le x \le \frac{\ell}{2}$ and $f(x) = k(\ell - x)$ f

$\frac{\ell}{2} \le x \le \ell$.

(d) The following table gives the variations

periodic current over a period.

t(sec) :	0	T/6	T/3	T/2	2T/3	5T/6	T
A(amp) :	1.98	1.30	1.05	1.30	–0.88	–.25	1.98

Show that there is a direct current part o

0.75 amp in the variable current and obtai

the amplitude of the first harmonic.

3.

Q. 2. (a) State Laplace transform of any periodic

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function. 2

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(b) Find Laplace transform of following functions :

(i) $\frac{1-\cos t}{t^2}$ 4

(ii) $\frac{e^{-at}-e^{-bt}}{t}$ 3

(c) (i) Find inverse Laplace transform of 4

$$\frac{s}{s^4+s^2+1}$$

(ii) Using convolution theorem, evaluate : 3

$$L^{-1}\left[\frac{1}{s(s^2+4)}\right]$$

(d) Solve $\frac{d^2x}{dt^2}+9x=\cos 2t$ if $x(0) = 1$, $x\left(\pi/2\right)=-1$. 7

3. (a) What is the Charpit's subsidiary equation. 2

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(b) Solve the equation :

$$px\,(z-2y^2) = (z-9y)\,(z-y^2-2x^3)$$

(c) Solve the equation :

$$(D^2 - DD' - 2D'^2)z = (y-1)e^x$$

(d) Solve the equation by method of separat of variables :

$4\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}=3u$ given $u = 3e^{-y} - e^{-5y}$ wl $x = 0$.

Q. 4. (a) Write condition for an analytic function.

(b) If f(z) is holomorphic function of z, sh that :

$$\left\{\frac{\partial}{\partial x}|f(z)|\right\}^2+\left\{\frac{\partial}{\partial y}|f(z)|\right\}^2=|f'(z)|^2$$

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(c) (i) Find the Laurent's series expansion of $\frac{z^2-1}{z^2+5z+6}$ about $z = 0$ in the region $2 < |z| < 3$. 4

(ii) Evaluate $\int_c \frac{(z+4)}{z^2+2z+5}dz$ where C is $|z + 1 - i| = 2$. 3

(d) Apply the calculus of residues, to prove that : 7

$$\int_0^\infty \frac{x\sin x}{x^2+a^2}dx = \frac{\pi}{2}e^{-a}$$

Q. 5. (a) State initial & final value theorem. 2

(b) Find the z-transform of the following sequences :

(i) $z\left[\frac{a^n e^{-a}}{\lfloor n}\right]$ 3

(ii) $z\left[e^{-an}\sin n\theta\right] = \frac{z\,e^a \sin\theta}{z^2e^{2a} - 2ze^a\cos\theta + 1}$ 4

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(c) (i) Obtain $z^{-1}\left[\frac{1}{(z-2)(z-3)}\right]$ for 3

$|z| < 2$, $2 < |z| < 3$ & $|z| > 3$

(ii) Using inversion integrals find the inverse z-transform of the : 4

$$\frac{(z^2+z)}{(z-1)(z^2+1)}$$

(d) Solve the equation by using z-transforms $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = y_1 = 0$. 7

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